SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 2: Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	**RAVURU PRANATHI**
Register Number	**192525076**

COURSE OUTCOME COVERED
CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3) **Total Marks: 30**
Question Count: 3

Code of Conduct
I **R.Pranathi(192525076)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: R. Pranathi

Assessment Tasks - Case Study

1. **Case Study 1: Smart Retail Inventory Management System**
A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. **Case Study 2: Cloud-Based Document Collaboration System**
An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices.
Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. **Case Study 3: Online Food Ordering Platform**
A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced

through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices. Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

ANS:

A Smart Retail Inventory Management System is a cloud-based solution that helps retail companies manage their products, sales and inventory more efficiently. In this system, all retail stores are connected to a centralized cloud platform. Every store uses POS (Point of Sale) systems and dashboards to record customer purchases and update inventory information. These devices communicate with the cloud through REST Web APIs. The APIs help different devices exchange information quickly and accurately without any delay.

Cloud storage is one of the most important parts of this system. It stores product details, stock information, transaction records and customer purchase history in a secure location. Since the data is stored in the cloud, every store can access the latest inventory information in real time. If one product is sold in one branch, the stock quantity is automatically updated for all other branches. This prevents problems like selling products that are already out of stock and improves inventory accuracy.

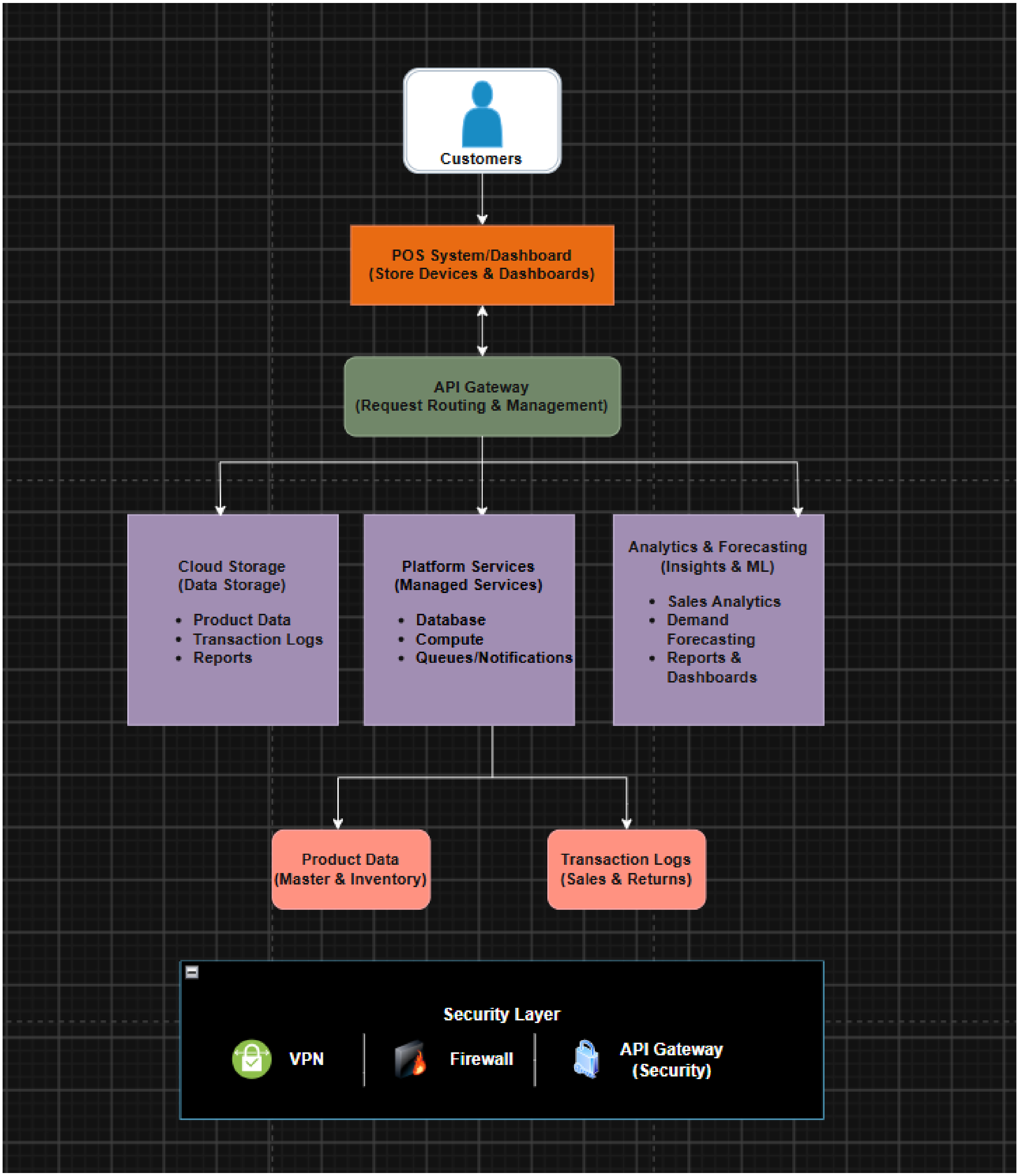
The system also uses cloud platform services for analytics and demand forecasting. Analytics services collect and analyze sales information from all stores. By studying previous sales records and customer buying patterns, the system predicts future product demand. For example, if the sales of umbrellas increase during the rainy season, the system predicts higher future demand and recommends ordering more umbrellas before the season begins. This helps the company avoid product shortages and reduces unnecessary storage costs.

Security is another important feature of this cloud solution. Secure communication between stores and the cloud is maintained using VPNs, which create a protected connection over the internet. Firewalls prevent unauthorized users from accessing the network. API Gateways manage all API requests and provide authentication before allowing access to cloud services. These security measures protect customer information and business data from cyber attacks.

Using cloud computing provides many advantages for the retail company. The system offers real-time inventory updates, centralized data management, automatic backups and easy scalability. If the company opens new branches, they can easily connect to the existing cloud platform without purchasing expensive hardware. Cloud services also reduce maintenance costs because software updates and server management are handled by the cloud provider.

For example, a retail company like Reliance Smart or D-Mart can use this system to manage inventory across hundreds of stores. When a customer purchases a product in Chennai, the inventory database is updated immediately. Managers can monitor stock levels using dashboards, while analytics services generate reports and predict future demand. This improves operational efficiency and customer satisfaction.

In conclusion, the Smart Retail Inventory Management System combines cloud storage, Web APIs, platform services and cloud security to provide an efficient, secure and scalable inventory solution. It reduces operational costs, improves inventory accuracy and helps businesses make better decisions using real-time analytics.


Customers
POS System/Dashboard
(Store Devices & Dashboards)
API Gateway
(Request Routing & Management)
Cloud Storage
(Data Storage)
• Product Data
• Transaction Logs
• Reports
Platform Services
(Managed Services)
• Database
• Compute
• Queues/Notifications
Analytics & Forecasting
(Insights & ML)
• Sales Analytics
• Demand Forecasting
• Reports & Dashboards
Product Data
(Master & Inventory)
Transaction Logs
(Sales & Returns)
Security Layer
VPN
Firewall
API Gateway
(Security)

2. Case Study 2: Cloud-Based Document Collaboration System
An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices.
Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

ANS:

A Cloud-Based Document Collaboration System allows multiple users to create, edit and share documents over the internet. Users access the application through web browsers or mobile devices without installing special software. This system works similarly to Google Docs, where many users can work on the same document at the same time from different locations.

The documents are stored in distributed cloud storage instead of local computers. Distributed storage keeps multiple copies of the same document on different servers. If one server fails, another copy is available, ensuring that users never lose their important files. Cloud storage also provides automatic backup and high availability so that documents remain accessible at any time.

An important feature of this system is version control. Every time a user edits a document, the system saves the changes as a new version. If a mistake is made or important information is deleted accidentally, users can restore a previous version of the document. Version control also records who made each change, making teamwork easier and more organized.

Web APIs are responsible for synchronizing documents across different devices. Suppose one employee edits a document on a laptop while another employee opens the same document on a mobile phone. The changes appear immediately because the APIs continuously synchronize the document with the cloud server. This provides real-time collaboration without conflicts.

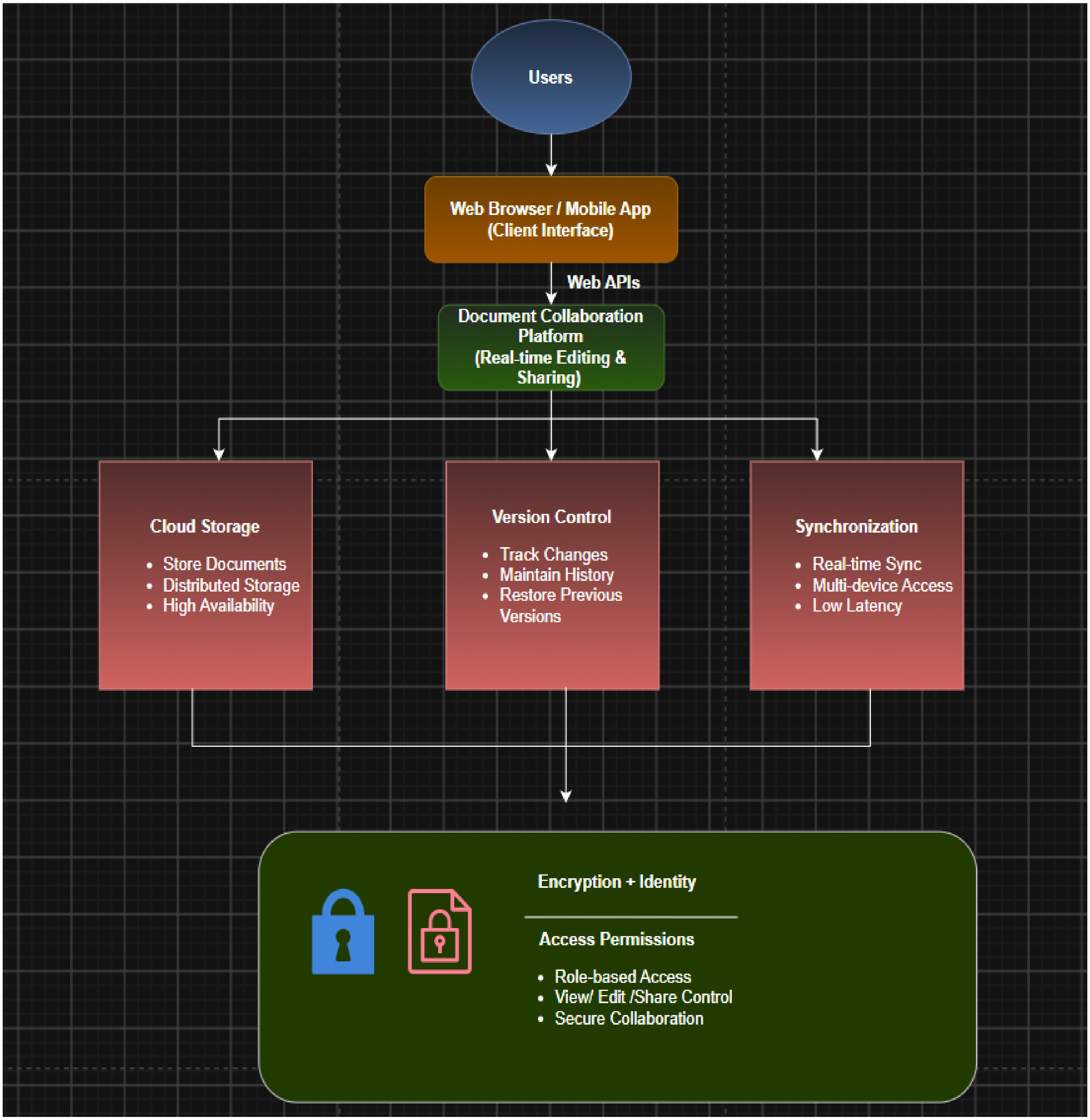
Cloud infrastructure ensures low latency and high availability. Cloud providers have data centers in many countries, allowing users to connect to the nearest server. This reduces delay and improves the speed of document access. Even if one data center experiences a problem, another data center continues providing the service.

Security is very important because business documents often contain confidential information. Cloud providers protect documents using encryption while storing the files and during data transmission. Identity management verifies the identity of every user before granting access. Access permissions allow administrators to decide who can view, edit or share each document. This prevents unauthorized users from modifying sensitive information.

The system provides many benefits to organizations. Employees can work together from different locations, improving teamwork and productivity. There is no need to carry storage devices because documents are available through the cloud. Automatic backup, version control and secure access make document management easier and more reliable.

For example, a multinational company preparing a project report can allow employees from India, the USA and the UK to edit the same document simultaneously. Every change is saved automatically, synchronized instantly and protected by encryption. Managers can also review previous versions whenever necessary.

In conclusion, the Cloud-Based Document Collaboration System improves communication, increases productivity and provides secure document management through cloud storage, Web APIs and platform services.



3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

ANS:

An Online Food Ordering Platform is a cloud-based application that allows customers to order food using websites and mobile applications. Restaurants, delivery partners and customers are connected through a centralized cloud platform. The frontend application communicates with backend cloud services using RESTful Web APIs. These APIs receive customer requests, process orders and send responses quickly.

Cloud storage is used to securely store menu images, restaurant details, customer information, invoices and order history. Since the data is stored in the cloud, restaurants can update menu items at any time, and customers always see the latest information. Cloud storage also provides backup and protects important business data.

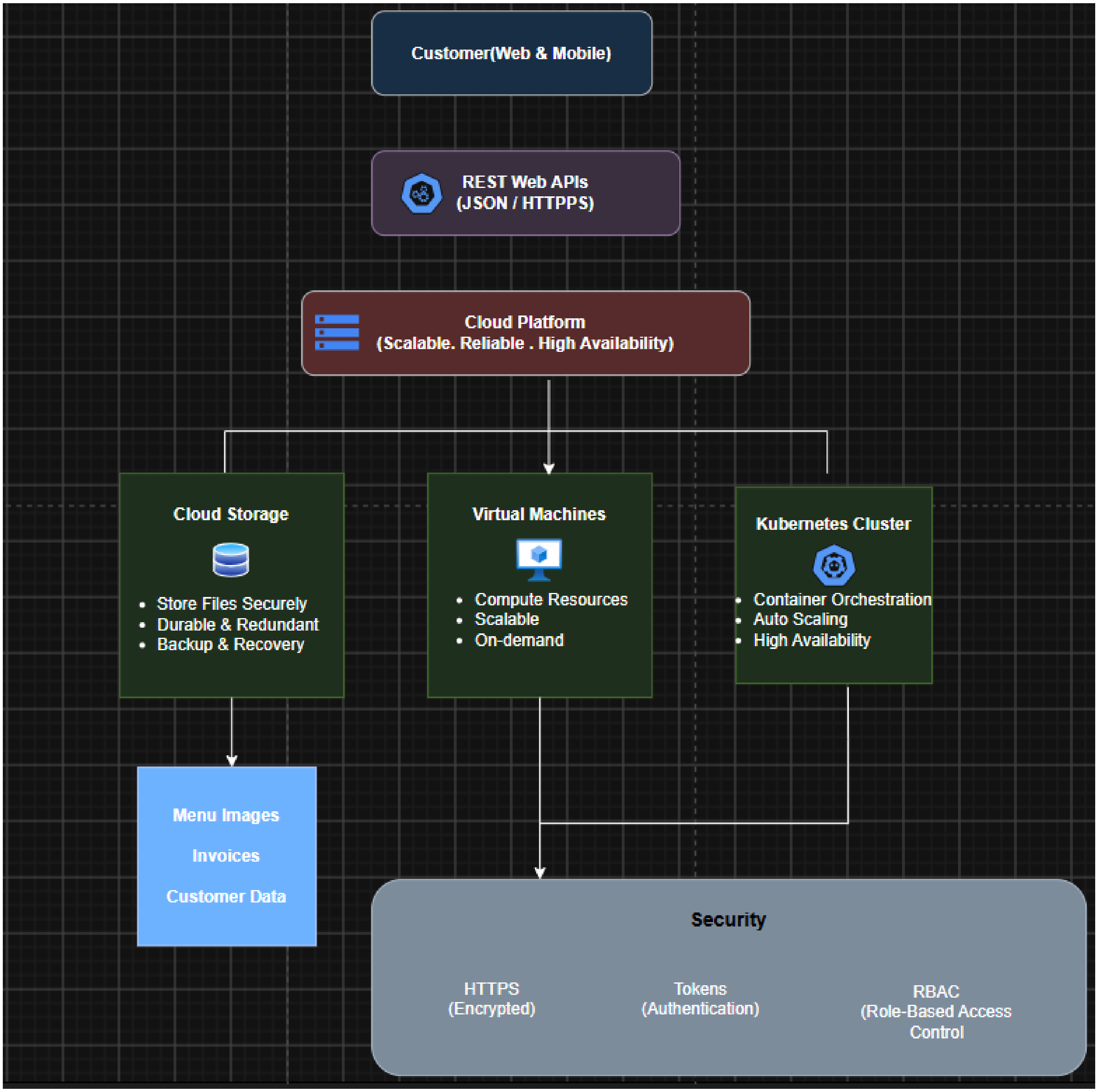
The application runs on Virtual Machines and managed Kubernetes services. Virtual Machines provide computing resources required to run the application. Kubernetes manages application containers automatically by distributing traffic, restarting failed containers and increasing the number of containers during peak hours. This ensures that the application continues to work smoothly even when thousands of customers place orders simultaneously.

Security is an important part of the system. HTTPS encrypts communication between customers and the server to prevent data theft. Authentication tokens verify the identity of users before allowing access to their accounts. Role-Based Access Control (RBAC) assigns different permissions to customers, restaurant owners, delivery partners and administrators. Each user can access only the information related to their role.

Cloud computing provides many advantages to this platform. It supports automatic scaling during busy hours such as weekends and festivals. It offers high availability because cloud services continue running even if one server fails. Cloud infrastructure also reduces maintenance costs since hardware management is handled by the cloud provider.

For example, platforms like Swiggy or Zomato receive thousands of food orders every minute. When customers place orders using mobile applications, REST APIs send the request to cloud servers. Kubernetes automatically distributes the workload among different servers, while cloud storage saves invoices and customer information securely. Authentication and HTTPS protect payment details and customer accounts.

In conclusion, the Online Food Ordering Platform successfully integrates Web APIs, cloud storage, Virtual Machines, Kubernetes and cloud security to provide a reliable, scalable and secure food delivery service. It improves customer experience, supports business growth and ensures continuous service even during high traffic periods.


Customer(Web & Mobile)
REST Web APIs
(JSON / HTTPPS)
Cloud Platform
(Scalable. Reliable . High Availability)
Cloud Storage
Store Files Securely
Durable & Redundant
Backup & Recovery
Virtual Machines
Compute Resources
Scalable
On-demand
Kubernetes Cluster
Container Orchestration
Auto Scaling
High Availability
Menu Images
Invoices
Customer Data
Security
HTTPS
(Encrypted)
Tokens
(Authentication)
RBAC
(Role-Based Access
Control